BAR ORYAN

Assistant Professor
Department of Earth and Environmental Sciences
Lehigh University

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APPOINTMENTS

Assistant Professor, Department of Earth and Environmental Sciences — Jul 2026 – present
Lehigh University, Bethlehem, Pennsylvania, USA

Green Postdoctoral Fellow, Institute of Geophysics and Planetary Physics — Jan 2024 – Jun 2026
Scripps Institution of Oceanography, La Jolla, California, USA
Advisor: Prof. Alice-Agnes Gabriel

Postdoctoral Research Scientist, Laboratoire de Géologie — Mar 2022 – Dec 2023
École Normale Supérieure, Paris, France
Advisors: Dr. Jean-Arthur Olive; Prof. Romain Jolivet

EDUCATION

Ph.D., Geophysics — Sep 2016 – Feb 2022
Lamont-Doherty Earth Observatory, Columbia University, New York, USA
Advisors: Prof. W. Roger Buck; Prof. Michael S. Steckler

M.Sc., Geophysics — Jul 2014 – Jul 2016
Tel Aviv University, Tel Aviv, Israel

B.Sc., Physics and Geosciences — Oct 2010 – Jun 2014
Tel Aviv University, Tel Aviv, Israel

PUBLICATIONS

15. **Oryan, B.**, Gabriel, A.-A., Bürgmann, R., Calais, E., Cheng, G., Chlieh, M., … Yu, S.-B. (2026). The Coupling Cloud: A community database of megathrust kinematic coupling models. *Seismica*, 5(1). https://doi.org/10.26443/seismica.v5i1.2314

14. Gabriel, A.-A., Karki, P., Magen, Y., **Oryan, B.**, Ulrich, T., Yun, J., & May, D. A. (2026). Tandem: An open-source high-performance computing volumetric software package to model sequences of earthquakes and aseismic slip across complex fault systems. *Seismological Research Letters*, 1–18. https://doi.org/10.1785/0220250436

13. Rao, M. P., Pacheco-Solana, A., Li, R., **Oryan, B.**, Jensen, J. E., Rodriguez-Caton, M., … Magney, T. S. (2026). Decoupled carbon assimilation and growth responses to aridity in temperate deciduous oaks. *Science Advances*, 12(24), eady7139. https://doi.org/10.1126/sciadv.ady7139

12. **Oryan, B.**, Gailleton, B., Olive, J.-A., Malatesta, L., & Jolivet, R. (2025). Inferring long-term tectonic uplift patterns from Bayesian inversion of fluvially-incised landscapes. *Journal of Geophysical Research: Solid Earth*. https://doi.org/10.1029/2024JB030819

11. Chong, J.-H., **Oryan, B.**, Shen, L., Steckler, M. S., & Lindsey, E. O. (2025). Interseismic uplift of anticlines above the Rakhine-Bangladesh megathrust from ALOS-2 InSAR. *Journal of Geophysical Research: Solid Earth*. https://doi.org/10.1029/2024JB030003

10. **Oryan, B.**, Olive, J.-A., Jolivet, R., Malatesta, L. C., Gailleton, B., & Bruhat, L. (2024). Megathrust locking encoded in subduction landscapes. *Science Advances*, 10(17), eadl4286. https://doi.org/10.1126/sciadv.adl4286

9. Kutschera, F., Jia, Z., **Oryan, B.**, Wong, J. W. C., Fan, W., & Gabriel, A.-A. (2024). The multi-segment complexity of the 2024 Mw 7.5 Noto Peninsula earthquake governs tsunami generation. *Geophysical Research Letters*, 51(21), e2024GL109790.

8. Steckler, M. S., Jaman, M. H., Grall, C. J., Goodbred, S. L., Wilson, C. A., & **Oryan, B.** (2024). Contribution of campaign GNSS toward parsing subsidence rates by time and depth in coastal Bangladesh. *Frontiers in Earth Science*, 12, 1354686.

7. **Oryan, B.**, Betka, P. M., Steckler, M. S., Nooner, S. L., Lindsey, E. O., Mondal, D., … Than, O. (2023). New GNSS and geological data from the Indo-Burman subduction zone indicate active convergence on both a locked megathrust and the Kabaw Fault. *Journal of Geophysical Research: Solid Earth*, 128(4), e2022JB025550.

6. Steckler, M. S., **Oryan, B.**, Wilson, C. A., Grall, C., Nooner, S. L., Mondal, D. R., … Goodbred, S. L. (2022). Synthesis of the distribution of subsidence of the lower Ganges-Brahmaputra Delta, Bangladesh. *Earth-Science Reviews*, 224, 103887.

5. **Oryan, B.**, & Savage, H. (2021). Regional heat flow analysis reveals frictionally weak Dead Sea fault. *Geochemistry, Geophysics, Geosystems*, 22(12), e2021GC010115.

4. **Oryan, B.**, & Buck, W. R. (2020). Larger tsunamis from megathrust earthquakes where slab dip is reduced. *Nature Geoscience*, 13, 319–324.

3. **Oryan, B.**, Villinger, H., Lazar, M., Schwab, M. J., Neugebauer, I., & Ben-Avraham, Z. (2019). Heat flow in the Dead Sea from the ICDP boreholes and its implication for the structure of the basin. *Quaternary Science Reviews*, 210, 103–112.

2. Malinverno, A., Cook, A. E., Daigle, H., & **Oryan, B.** (2018). Glacial cycles influence marine methane hydrate formation. *Geophysical Research Letters*, 45(2), 724–732.

UNDER REVIEW

1. **Oryan, B.**, & Gabriel, A.-A. Do coupled megathrusts rupture? *Preprint*. https://doi.org/10.31223/X5HB3N

GRANTS & AWARDS

SCEC Workshop Grant ($20,000) 2025
Tandem Workshop: Modeling the Earthquake Cycle from Faults to Supercomputers. Oryan, B., Gabriel, A.-A., May, D., Yun, J., Magen, Y.

CRESCENT Geoscience Professional Development Fellowship 2025

Cecil H. and Ida M. Green Foundation for Earth Sciences ($3,000) 2025
Tandem Supercomputing Hackathon. Oryan, B., Gabriel, A.-A., May, D.

Cecil H. and Ida M. Green Foundation for Earth Sciences ($2,000) 2024
Seismic Cycle and Aseismic Slip Supercomputing Hackathon. Oryan, B., Gabriel, A.-A., May, D.

CRESCENT Seed Grant ($30,000) 2024
Synergizing megathrust seismo-geodetic coupling and slip models using optimal transport and machine learning frameworks to mitigate earthquake hazard in Cascadia. Oryan, B., Gabriel, A.-A.

Top 10% Most-Read Author, *Journal of Geophysical Research: Solid Earth* 2023

IGPP Green Postdoctoral Fellowship ($149,000) 2023

NASA Earth Surface and Interior Grant ($650,000) 2022
GNSS and InSAR surface deformation constraints on a 3-D geodetic model of the Indo-Burma subduction zone. Steckler, M., Lindsey, E., Oryan, B., et al.

AGU Outstanding Student Presentation Award 2021

Lamont-Doherty Earth Observatory Climate Center Grant ($10,000) 2021
Temporal dynamics of tree growth and photosynthesis and their environmental drivers in the Lamont Sanctuary Forest Preserve. Rao, M., Pacheco-Solana, B., Oryan, B., et al.

Chevron Student Incentive Fund ($3,200) 2020
Developing the LDEO PhenoCam network to track the fate of forest carbon from photosynthesis to growth. Oryan, B., Rao, M.

Chateaubriand Fellowship ($6,000) 2020
Financial support to work with Dr. Jean-Arthur Olive at the Laboratoire de Géologie, École Normale Supérieure.

Stork Fund ($12,500) 2019
Department of Earth and Environmental Sciences graduate student field trip to Peru. Oryan, B., Myers, E.

Dean's Fellow, Department of Earth and Environmental Sciences, Columbia University 2016

M.Sc. Excellence Scholarship, Tel Aviv University ($1,000) 2015

PRESENTATIONS

University of Arizona, Department of Geosciences (invited talk) Mar 2026

Rice University, Department of Earth, Environmental and Planetary Sciences (invited talk) Feb 2026

AGU Fall Meeting (talk) Dec 2025
Locked but not loaded? A global analysis of megathrust coupling and earthquake slip.

University of Nevada, Reno, Department of Geological Sciences and Engineering (invited talk) Nov 2025

SSA Annual Meeting (talk) Apr 2025
Rock, river, record: Reading the geomorphic record of seismic cycles through Bayesian inversion of river-incised landscapes.

Rutgers University, Department of Earth and Planetary Sciences (invited talk) Mar 2025

Dartmouth College, Department of Earth Sciences (invited talk) Feb 2025

University of Oregon, Department of Earth Sciences (invited talk) Jan 2025

AGU Fall Meeting (invited talk) Dec 2024
Subduction sculpting when megathrusts sleep: How interseismic deformation encodes a signature of plate locking in forearc landscapes.

AGU Fall Meeting (talk) Dec 2024
Inferring spatially-variable rock uplift from fluvially-incised landscapes: A Bayesian inversion framework in χ-space.

IRN-Andes-FRENSZ Seminar (invited talk) Oct 2024
Megathrust locking encoded in subduction landscapes.

AGU Fall Meeting (talk) Dec 2022
Spatial patterns of long-term forearc uplift inferred from river profiles, and their comparison with short-term deformation.

AGU Fall Meeting (talk) Dec 2022
New GNSS and geological data from the Indo-Burman subduction zone indicate active convergence on both a locked megathrust and the Kabaw Fault.

JpGU Meeting (invited talk) May 2022

EGU General Assembly (talk) May 2022

AGU Fall Meeting (invited talk) Dec 2021

Caltech Seismological Laboratory Seminar (invited talk) Oct 2021

TEACHING

Co-Mentor, Heat Advection Experiment, École Normale Supérieure — Fall 2022
Guided students in conducting and analyzing a heat advection experiment.

Instructor, Field Trip to Peru, Columbia University — 2019
Led and organized a geological field trip to Peru.

Mentor, LDEO Summer Intern Program — Summer 2019
Mentored a group of high school students from low-income backgrounds.

Teaching Assistant, Life Systems, Columbia University — Spring 2019

Teaching Assistant, Geodynamics, Columbia University — Fall 2018

Teaching Assistant, Lab in Geosciences, Tel Aviv University — Fall 2015
Designed and developed a lab experiment emulating the heat flow of the Earth for an undergraduate class.

Tutor, AP Calculus and Electromagnetism, Tel Aviv University — Fall 2014

FIELD WORK

HT-RESIST Electromagnetic Research Cruise, New Zealand — Winter 2019
Deployment and recovery of 120 EM receivers and 500 line-km of EM source.

Borehole Temperature Profile Measurements, Dead Sea, Israel — Summer 2015
ICDP Dead Sea borehole temperature measurements.

Thermal Conductivity Measurements, IODP Core Repository, Bremen, Germany — Summer 2014
Thermal conductivity measurements of ICDP Dead Sea cores using a KD2 Pro probe and optical apparatus.

OUTREACH

Venn Plate Tectonics Happy Hour — 2023 – 2025

Science Writer, Little Big Science NGO — 2018 – 2025

Research as Art Committee, Lamont-Doherty Earth Observatory — 2018 – 2019

Open House, Lamont-Doherty Earth Observatory — 2016 – 2021

Earth-Sun Day, American Museum of Natural History — 2017 – 2019